BARSHAN GHOSH

Computer Science Graduate | Software & IT Systems | Applied AI | Technology Education

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Portfolio | All GitHub Repositories | GitHub Profile | LinkedIn

PROFESSIONAL SUMMARY

Computer Science and Engineering graduate (CGPA 3.83/4.00) with experience across IT operations and consulting, web/software projects, applied AI/ML learning, digital marketing, and technology education. Builds practical projects using modern web technologies and database-backed systems, with a particular focus on data integrity, workflow automation, and usable tools for real-world problems. Also experienced in school IT coordination, computer instruction, and private tutoring.

TECHNICAL SKILLS

Web & Software: Next.js, React, Astro, TypeScript, JavaScript, HTML/CSS, Tailwind CSS, Node.js, REST APIs

Databases & Platforms: PostgreSQL, Supabase, MySQL, Firebase, SQL Server, WordPress, Shopify

AI, Data & Testing: Python, Weka, TensorFlow/Keras fundamentals, Power BI, Vitest, Playwright, Kaggle

Systems & Tools: Git/GitHub, Vercel, Linux, Windows, hardware and network troubleshooting, Docker basics

Digital & Education Tools: SEO, GA4, Google Tag Manager, Meta Pixel, Google Search Console, Canva, Google Workspace, MS Office, Google Classroom, Zoom, Microsoft Teams

Additional Programming: C, C++, Java; PHP (basic reading and modification)

SELECTED PROJECTS

Money Calculator | Next.js, TypeScript, Supabase, PostgreSQL, Vitest, Playwright

In-progress personal-finance application built around ledger-backed records. Includes transaction correction/reversal history, account balances, lending and borrowing lifecycle, tuition payments, asset tracking, budgeting, reporting, and financial forecasts. Repository includes unit and database-integration testing for key financial invariants.

View source repository

Durga-Map | Android, GPS, Maps, Capacitor

Android-focused Durga Puja location explorer that extracts GPS coordinates from photos, organizes location records, and helps users open mapped locations in Google Maps. Includes photo-metadata handling and an Android app workflow.

View source repository

ASTA Store | Astro, TypeScript, Supabase

E-commerce storefront for handmade leather bags with product browsing, cart management, checkout flow, Supabase authentication, and an administrative product-management area.

View source repository

Mount Royal Academy Website | Astro, TypeScript, Supabase

School website project with admissions/application flow, academic calendar, notices, gallery management, and authenticated content administration.

View source repository

999 National Emergency Service | C#, Windows Forms, SQL Server

Academic desktop application concept for citizen registration, emergency-service request entry with time/location details, and administrative request monitoring. Not a live emergency dispatch service.

View source repository

Blockchain Internet of Things (bIoT) Thesis | Research

Undergraduate literature review and proposed architecture exploring blockchain-based security for healthcare IoT, including authentication, data integrity, remote patient monitoring, and medical-data privacy.

View source repository

Student Performance Dashboard | Analytics & Data Visualization

Dashboard concept for presenting student results, attendance, and classroom-performance indicators using illustrative sample data; not connected to a live school database.

View source repository

Digital Coaching Spreadsheet Automation | Google Apps Script

Small workflow automation that adds a missing current-date column to an Attendance sheet and a missing current-month column to a Payment sheet, avoiding duplicate headers.

View source repository

PROFESSIONAL EXPERIENCE

IT Coordinator | Computer Teacher | O-Level Physics Guest Lecturer | *Jun 2025 – Present*

Mount Royal Academy — Narayanganj, Bangladesh

- Support day-to-day school IT infrastructure and technology operations.
- Teach computer/ICT concepts to students in Grades 3–7.
- Deliver guest instruction in O-Level Physics.

Proxy ICT Teacher | *Apr 2025 – May 2025*

Heritage School — Narayanganj, Bangladesh

- Covered ICT classes during regular teachers' absences and maintained continuity with planned lessons.
- Led classroom technology activities and supported student learning.

Digital Marketer (Remote) | *Oct 2024 – Jan 2025*

Edupy Academy — Dhaka, Bangladesh

- Supported digital visibility, social-media outreach, and promotional content across online channels.

IT Associate (Hybrid) | *Jan 2023 – Sep 2024*

CNBDEXIM — Dhaka, Bangladesh

- Maintained day-to-day IT systems and assisted with hardware support and network troubleshooting.
- Provided on-site and remote technical assistance for business operations.

IT Consultant (Remote) | *Aug 2022 – Oct 2024*

Digital Fire Safety Solution Ltd — Dhaka, Bangladesh

- Provided remote technical consulting and troubleshooting for operational IT issues.
- Reviewed IT needs and recommended practical improvements to system reliability and efficiency.

AI Engineer Intern (Remote) | *Aug 2022 – Jan 2023*

Teachnook — India

- Participated in AI/ML learning and project activities involving model-building, training, and testing workflows.
- Completed a cancer-prediction learning exercise using a Kaggle dataset and tools including Weka, Excel, and Google Colab.

EDUCATION

B.Sc. in Computer Science and Engineering | 2019 – 2023

American International University-Bangladesh (AIUB) — CGPA: 3.83/4.00; completed in 3 years

Higher Secondary Certificate (Science) | 2017 – 2019

Government Tolaram College — GPA: 3.92/5.00

Secondary School Certificate (Science) | 2014 – 2017

Narayanganj High & College — GPA: 5.00/5.00

PRIVATE TUTORING EXPERIENCE

Private Tutor — O-Level / Edexcel (Jun 2026) — Physics, ICT, Computer Science, Mathematics B, Further Pure Mathematics

Private Tutor — HSC Candidate (Science) (Oct 2024) — ICT, English, Bengali

Private Tutor — SSC Candidate (Arts) (Jan 2023) — ICT, English, General Mathematics, General Science

Private Tutor — SSC Candidate (Science) (Jan 2023) — Physics, Chemistry, Higher Mathematics, Biology

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Operating System — Coursera | Verify credential
- AI for Everyone — Coursera | Verify credential
- Python for Everybody — Coursera Specialization | Verify credential
- IT Fundamentals for Cybersecurity — Coursera Specialization | Verify credential
- MS Excel: Beginner to Advanced | Verify credential
- MS PowerPoint Advanced Course | Verify credential
- Professional Digital Marketing (A+) — Creative IT | Verify credential
- Digital Marketing Business Workshop | Verify credential

LANGUAGES

Bengali (native) | English | Hindi